

Carl Emmanuel M. Macabales

0956 389 3104 | carlmacabales31@gmail.com | cemmacabales.tech | [linkedin/cemmacabales](https://linkedin.com/in/cemmacabales)

EDUCATION

Mapúa University

Bachelor of Science in Computer Science, Specializing in Artificial Intelligence

Makati, Philippines

Aug. 2023 – Jul. 2026

EXPERIENCE

The BLOKC (BLOCKLABS Inc.)

Software Engineering Intern

Apr. – Jun. 2026

- Led a **5-intern** engineering pod across three products, standardizing code review and test workflows to cut average feature turnaround from **4 weeks to 2 weeks** while lowering post-release error rate by **~40%**.
- Ran security-focused QA for a Learning Management System, filing **30+** reproducible issues covering IDOR, broken access control, and rate-limit bypass, cutting the pre-release defect escape rate by **~35%**.
- Built a multi-provider LLM adapter (**Groq, Anthropic, Gemini, Vertex AI, Ollama**) for Paiflow that turns prompts into Stellar payment flows, lifting flow-generation success to **~94%** and cutting malformed-output crashes by **~70%**.
- Shipped a **6-phase** on-chain payout system with atomic withdrawals and anti-fraud controls, stress-testing **10K+** simulated payouts on Stellar testnet with **zero** double-spends and **5/5** supervisor ratings.

PUBLICATIONS

Multi-Class Kidney Abnormality Segmentation

Aug. 2025 – Jan. 2026

AI - Deep Learning Computer Vision | ICIPCN 2026

- Developed a YOLOv12 model to detect kidney cysts, stones, and tumors from CT scans.
- Curated and augmented a **14,761-image** CT dataset to improve generalization.
- Achieved **mAP@0.5 of 0.946** (axial) and **0.885** (coronal), reducing radiologist workload.
- Presented at the 6th ICIPCN, Kathmandu University, Nepal.

RAG-Based Clinical Guideline Chatbot for Atrial Fibrillation

Dec. 2025 – Mar. 2026

AI – Natural Language Processing | CSPA 2026

- Built a RAG chatbot (**Llama-3, Phi-3, Qwen3**) for querying **100+ pages** of ESC AF guidelines.
- Improved retrieval with MedCPT, FAISS, and BGE reranking, reaching **BERTScore F1 0.835** and **ROUGE-1 0.456**.
- Evaluated on **20** clinical queries, hitting faithfulness up to **8.75/10** at **~0.7s** latency and **7.9 tok/s**.

PROJECTS

Real-Time AI Exercise Coaching System

Apr. – Jun. 2026

AI – Computer Vision | Python, MediaPipe BlazePose, TFLite, ONNX, Raspberry Pi 5

- Trained a dual-head LSTM on **451,638** windows from **174 videos**, reaching **97.15%** exercise and **92.81%** form accuracy across **9 movements**.
- Exported to a **219 KB** TFLite model with BlazePose Lite, hitting **25–30 FPS** on a Raspberry Pi 5.
- Built a **~115 MB** on-device RAG chatbot over PDF fitness manuals, removing PyTorch from the Pi runtime.

ACTIVITIES

Pink Raft — 1st Runner-Up, Stellar Hackathon 2026

May 2026

Full-Stack Engineer & AI Integration Lead

- Built a no-code Stellar/Soroban payment flow builder that deploys live smart contracts in under **60 seconds**.
- Cut AI response latency **~80%** (10s → 2s) via schema hardening, and shipped WebAuthn 2FA with non-custodial wallet signing.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, SQL, HTML/CSS, R

Frameworks & Libraries: React, React Native, Flask, FastAPI, Expo, Node.js, SCSS

AI/ML: RAG, NLP, Computer Vision, Object Detection, PyTorch, OpenCV, NumPy, pandas

Developer Tools: Git, Docker, Firebase, Firestore, Streamlit, VS Code, NVIDIA CUDA

Cloud & Databases: Firebase, Appwrite, PostgreSQL, Google Cloud Platform, RESTful APIs